

Q-Write a short scenario where a user tries to access a private WiFi network (like at a campus or airport) and describe how each of the three AAA functions—Authentication, Authorization, and Accounting—would apply step by step in this situation.

Scenario: Accessing a Private Campus Wi-Fi Network

Imagine **Rahul is a student at a college** and wants to connect his laptop to the college's private Wi-Fi network.

1. Authentication – “Who are you?”

Rahul connects to the college Wi-Fi and enters his student username and password. The Wi-Fi system checks his credentials with the authentication server.

→ If the credentials are correct, Rahul's identity is verified.

2. Authorization – “What are you allowed to access?”

After authentication, the network checks Rahul's permissions. As a student, he may be allowed to access the internet and college websites, but not administrative servers.

→ The system gives Rahul only the network access he is permitted to use.

3. Accounting – “What did you do/use?”

The network records information such as Rahul's login time, logout time, IP address, and amount of data used.

→ This information can help the college monitor usage, troubleshoot problems, and maintain network security.